Exercise 1. Prove that for any non-negative integer n the number

$$3^{4^n} + 23$$

is divisible by 13.

Exercise 2. Determine the remainder when the number:

$$1! + 2! + 3! + 4! + \cdots + 100!$$

is divided by 12.

Exercise 3. Show that $2^{20} - 1$ is divisible by 41.